

LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Experienced in building performance-optimized, offline-first mobile applications with real-time ML capabilities and cloud-backed sync (Firebase, Supabase). Strong problem-solving skills and dedication to writing clean, maintainable code. Proven ability to bridge AI research and mobile deployment — from model training to TFLite on-device inference.

EDUCATION

Nha Trang University

Bachelor of Information Technology

- **GPA:** 3.25/4.0

2022 – 2026

Nha Trang, Vietnam

TECHNICAL SKILLS

Languages: Java, Kotlin, JavaScript, Python, SQL

Mobile: Android SDK, Jetpack Compose, CameraX, Room DB, WorkManager, Coroutines & Flow

AI / Computer Vision: TensorFlow Lite, MediaPipe, MLKit (Pose Estimation), YOLO, OpenCV, Gemini API, Groq API (LLaMA)

Backend Services: Firebase (Auth, Firestore, Storage), Supabase, REST API design, OkHttp

Tools: Git/GitHub, Android Studio, Gradle, ADB (Android Debug Bridge), KSP, VS Code

AI-Assisted Dev: Google Antigravity, OpenAI Codex

Languages: Vietnamese (Native) – English (Intermediate - proficient in technical documentation)

PROJECTS

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Developed an offline-first Android application from scratch using Kotlin, Jetpack Compose, MVVM, and clean code practices.
- Solved the problem of passive vocabulary memorization by implementing real-time on-device object detection (CameraX + TensorFlow Lite), integrating Gemini AI to generate contextual learning stories from scanned vocabulary.
- Addressed the “forgetting curve” problem in language learning by designing a custom SM-2 Spaced Repetition engine that automatically schedules reviews at optimal intervals, backed by a hybrid Room DB/Firestore sync system to ensure reliable offline data sync.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator, alongside a chess-style game review UI for match accuracy analysis and AI feedback.

Lung X-ray Trustworthy Classifier (Android + Research) | *Graduation Project, NTU*

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- **Tech Stack:** Python, TFLite, U-Net (ResNet50), YOLOv12m-cls, Kotlin, Jetpack Compose, Groq API.
- Deployed a 2-stage AI pipeline (U-Net segmentation → YOLOv12m-cls classification) to a native Android app via TFLite, enabling fully offline on-device inference across 4 pathology classes (COVID-19, Viral Pneumonia, Lung Opacity, Normal).
- Engineered median-fill masking to eliminate attention leakage outside lung regions, validated with EBPg (+12% vs. baseline) and IoU localization across 300 annotated cases.
- Built an OOD safety gate (lung-area ratio thresholding, 8–72% valid range) to auto-reject invalid inputs, and integrated dual-mode XAI (Occlusion Sensitivity + Grad-CAM++).
- Added a context-aware LLM assistant (Groq/LLaMA 3.1, Fleischner-guideline grounded) and native PDF report export, benchmarked against 5 backbone architectures.

AI Fitness Coach | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, CameraX, MediaPipe/MLKit (Pose Estimation), Room DB, Text-To-Speech (TTS).

- Developed an offline-first Android app (MVVM) providing a comprehensive ecosystem with an 800+ exercise library and an AI-driven personalized workout recommendation engine.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe) to extract 33 joint coordinates and calculate 3D vector angles in real-time (60 FPS).
- Built a custom rule-based feedback engine with Text-To-Speech integration (audio alerts) and automated rep counting, caching progress 100% offline via Room DB.